2020-Summer Paper 2 – Urology

Contents

1. Q1. Assessment and recent concepts in management of metastatic RCC (LAQ)
2. Q2. Surgical management of erectile dysfunction (LAQ)
3. Q3a. CRPC in Ca prostate (SAQ)
4. Q3b. Retractile testis (SAQ)
5. Q3c. Preoperative management of pheochromocytoma (SAQ)
6. Q3d. TRUS biopsy (SAQ)
7. Q3e. Management of valve bladder (SAQ)
8. Q3f. TOT (transobturator tape) (SAQ)

Q1. Describe assessment and recent concept in management of metastatic RCC.

Type: LAQ (25 marks)

Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Therapy moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

Assessment

- **Clinical:** cough, bone pain, neurological deficit, weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain imaging with neurological symptoms; PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.
- **Prognostic risk group** (decides first-line therapy):

Adverse factor	MSKCC (Memorial Sloan-Kettering), cytokine era	IMDC (International mRCC Database Consortium), TKI era
Karnofsky <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium	>10 mg/dL	> upper limit of normal (ULN)
LDH >1.5 × ULN	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to systemic therapy

Adverse factor	MSKCC (Memorial Sloan-Kettering), cytokine era	IMDC (International mRCC Database Consortium), TKI era
Neutrophils, platelets > ULN	No	Yes (two factors)

- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

Management: recent concepts

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

1. First-line therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	Alternatives
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib or pazopanib if IO (immuno-oncology) not possible; active surveillance for slow, low-volume disease
Intermediate/poor	The same three, **or nivolumab + ipilimumab**	Cabozantinib; sunitinib; pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing the brake on T cells and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). PD-L1 (programmed death ligand-1) status does not select patients.

2. Later lines: switch mechanism

- After IO: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor).
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS gain.

3. Cytoreductive nephrectomy (CN): routine to selective

- Cytokine era: SWOG 8949 and EORTC trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (OS 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; OS favoured deferral.
- **Now:** immediate CN for fit patients with low metastatic burden not needing immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.
- Palliative nephrectomy or angioembolisation for haematuria or pain.

4. Metastasectomy and local therapy

- Complete resection of solitary or few metastases: 5-year survival 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)**, radiosurgery for brain metastases, or ablation for oligometastatic disease; palliative radiotherapy or surgery for cord compression and weight-bearing bones.

5. Adjuvant immunotherapy (new)

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (metastases completely resected): disease-free survival HR 0.63; FDA (Food and Drug Administration) approval 2021; a later report showed OS benefit.

6. Supportive and special groups

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis) need steroids.
- Non-clear cell RCC: clinical trial preferred; cabozantinib or sunitinib; gemcitabine + carboplatin for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2): durable complete response 7–9%; temsirolimus for poor risk; chemotherapy and hormones are ineffective.

Recent advances

- Belzutifan, also approved for VHL-associated RCC; neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

Diagram to draw: flowchart: metastatic RCC → biopsy and IMDC risk → IO combination ± deferred CN → metastasectomy or SBRT → switch drug class on progression.

> **Key points to remember:**
>
> - IMDC: Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
> - IO/TKI for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
> - CARMENA and SURTIME: no routine upfront CN.
> - Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.

Q2. Discuss surgical management of erectile dysfunction.

Type: LAQ (25 marks)

Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- **Indications:** failed or unacceptable medical therapy (common after prostatectomy, pelvic radiotherapy, diabetes); cavernosal fibrosis (after ischaemic priapism, infection or implant removal); Peyronie disease with ED; penile or pelvic trauma.
- Detailed vascular testing is needed only before vascular surgery.

A. Penile vascular surgery

	Arterial revascularisation	Venous surgery
Defect	Arteriogenic (poor inflow)	Veno-occlusive (venous leak)
Work-up	Duplex; selective pudendal angiography (also checks the inferior epigastric donor)	DICC (dynamic infusion cavernosometry and cavernosography) locates the leak
Technique	Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation)	Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication
Selection	Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma	Not defined
Status	Strictly selected men	Investigational
Complications	Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia	Glans numbness, shortening, curvature, skin necrosis

B. Penile prosthesis

Types

Type	Features	Best for
3-piece inflatable (AMS 700 CX/LGX, Titan)	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity with true flaccidity**	Most men
2-piece inflatable	No reservoir; poor flaccidity; discontinued	–
Malleable rods (Tactra, Genesis)	Simple, cheap, almost never fail; always firm	Poor manual dexterity; cost

Contraindications: active skin, urinary or systemic infection; uncontrolled diabetes; unrealistic expectations. In insensate men (spinal cord injury) use an inflatable device, because rods erode.

Consent: the erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

Preoperative preparation

- Treat phimosis and skin or fungal lesions; sterile urine; stop anticoagulants and aspirin.

- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

Operative steps (3-piece, penoscrotal approach)

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Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures →
corporotomy
→ dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13
mm)
→ goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation
→ close corporotomies → pump in dependent subdartos pouch → reservoir in space of Retzius
→ connect tubing → test inflation → irrigate → close in layers
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- **Approaches:** penoscrotal (most used; secures the pump; urethra at risk); infrapubic (reservoir placed under vision; dorsal nerve at risk); subcoronal or ventral penile for rods.
- Malleable rods: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Ectopic submuscular reservoir** after pelvic surgery or cystectomy; a lockout valve prevents autoinflation.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

Complications

- **Mechanical failure** (commonest): repair in about 15% at 5 years, 30% at 10 years.
- **Infection (1–3%)**: biofilm on the device. Antibiotic-coated devices halve it; no-touch technique gives <0.5%. Treatment: remove **all** parts. **Mulcahy salvage** (antiseptic washout and immediate new device) succeeds in 84%; delayed reimplantation loses length.
- Erosion or extrusion (catheterisation, rods in insensate men), reservoir erosion into bladder or bowel, glans necrosis (diabetes, smoking, radiation, previous implant), cylinder aneurysm.

Special situations: ischaemic priapism >36 hours (early implantation preserves length); transplant recipients (no extra infection); after phalloplasty (single cylinder, high erosion).

Outcome: patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

Diagram to draw: 3-piece prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.

Key points to remember:

- Prosthesis is the last line but gives 80–90% satisfaction.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Infected implant: remove all parts; Mulcahy salvage 84%.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Q3a. CRPC in Ca prostate.

Type: SAQ (10 marks)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
  - **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
  - **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Mechanisms

- Castration-resistant is **not** androgen-independent. The AR (androgen receptor) still drives growth: AR amplification, mutation, splice variants, and adrenal or intratumoural androgen synthesis (CYP17).
- Also bypass pathways (PI3K/AKT) and neuroendocrine change (low PSA, visceral disease).

Evaluation

- Testosterone, PSA, PSADT (PSA doubling time); CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Germline and tumour testing for HRR (homologous recombination repair) genes such as BRCA2.
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).

Management

- **Continue ADT (androgen deprivation therapy)**; try antiandrogen withdrawal.
- **nmCRPC** (non-metastatic), PSADT ≤10 months: apalutamide (SPARTAN), enzalutamide (PROSPER) or darolutamide (ARAMIS) improve metastasis-free and overall survival.
- **mCRPC** (metastatic):

Class	Agent	Key trial
ARPI (AR-pathway inhibitor)	Abiraterone 1000 mg + prednisone; enzalutamide 160 mg	COU-AA-301/302; AFFIRM, PREVAIL
Taxane	Docetaxel 75 mg/m² 3-weekly; cabazitaxel after docetaxel	TAX 327; TROPIC, CARD
PARP (poly-ADP-ribose polymerase) inhibitor (HRR mutation)	Olaparib ± abiraterone	PROfound; PROpel
Radioligand	Radium-223 (symptomatic bone-only); ^{177}Lu-PSMA-617	ALSYMPCA; VISION
Immunotherapy	Sipuleucel-T; pembrolizumab if MSI-high (microsatellite instability)	IMPACT

- **Sequencing:** switch mechanism; a second ARPI rarely works (cross-resistance).
- Neuroendocrine variant: platinum + etoposide.
- **Supportive:** denosumab 120 mg SC or zoledronic acid 4 mg IV every 4 weeks against skeletal events; radiotherapy for bone pain; cord compression needs dexamethasone, urgent MRI and radiotherapy ± decompression.

Diagram to draw: clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

Key points to remember:

- Testosterone <50 ng/dL with PSA or radiological progression.
- AR still drives CRPC; keep ADT going.
- nmCRPC: apalutamide, enzalutamide, darolutamide.
- mCRPC: ARPI, taxanes, PARP inhibitor, radium-223, ^{177}Lu-PSMA.

Q3b. Retractile testis?

Type: SAQ (10 marks)

Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

Testis	Finding	Action
Retractile	Reaches scrotal bottom and stays	Observe yearly
Gliding (unstable high scrotal)	Pulled down but springs back	Treat as undescended
Undescended (palpable)	Cannot reach mid-scrotum	Orchidopexy by 18 months
Ascending (acquired)	Previously scrotal, now not	Orchidopexy

Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

Diagram to draw: inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

Q3c. Pre operative management of pheochromocytoma.

Type: SAQ (10 marks)

Why prepare

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

Step 1: Confirm, localise, test genes

- Plasma free or 24-hour urinary fractionated **metanephrines**; CT or MRI; ^{68}Ga-DOTATATE PET or MIBG (metaiodobenzylguanidine) for extra-adrenal or hereditary disease. **Never biopsy.**
- Genetic testing: about one-third hereditary (MEN2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau], NF1 [neurofibromatosis type 1], SDHB/SDHD [succinate dehydrogenase subunits]).

Step 2: Cardiac assessment

- ECG and echocardiography (catecholamine cardiomyopathy); joint plan with the anaesthetist.

Step 3: Drug blockade (7–14 days)

Drug	Action	Points
Phenoxybenzamine (first line)	Irreversible non-selective α-blocker	10 mg twice daily, increase every 2–3 days; about 1 mg/kg total
Doxazosin, prazosin	Selective α1-blockers	Fewer side effects
β1-blocker (atenolol, metoprolol)	Controls tachycardia	**Only after adequate α-blockade**
Calcium channel blocker	Smooth muscle relaxation	Alternative or add-on
Metyrosine	Blocks tyrosine hydroxylase	Add-on; sedation, extrapyramidal effects

- **Why α before β:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed: crisis.
- **Target:** seated BP 120–130/80 mmHg; mild postural drop (systolic >90 mmHg) shows adequate blockade.

Step 4: Volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV fluid the night before. This prevents post-excision hypotension.

Step 5: Day of surgery

- Last phenoxybenzamine dose the night before; omit the morning dose.
- Arterial and central lines; nitroprusside, nicardipine or phentolamine for surges; esmolol for tachyarrhythmia; early adrenal vein ligation.

Postoperative

- Fluids ± vasopressors for hypotension; glucose monitoring (rebound hypoglycaemia); HDU (high dependency unit) overnight; steroid cover after bilateral adrenalectomy.
- Metanephrines after surgery, then yearly for life (recurrence up to 16%).

Diagram to draw: flowchart: metanephrines → imaging → α-blockade → β-blockade → salt and fluids → surgery.

> **Key points to remember:**
> - α-blockade 7–14 days for all.
> - Never β before α.
> - Volume expansion prevents hypotension; watch glucose.

Q3d. TRUS biopsy.

Type: SAQ (10 marks)

Introduction

- TRUS (transrectal ultrasound)-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

Indications and contraindications

- Raised PSA (prostate-specific antigen) with risk factors; suspicious DRE (digital rectal examination); MRI PIRADS (Prostate Imaging Reporting and Data System) ≥3. Repeat: rising PSA, ASAP (atypical small acinar proliferation), active surveillance.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

Preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Antibiotics for all** (contaminated route): fluoroquinolone, or cephalosporin ± aminoglycoside, ≤24 hours. Augmented or swab-directed if risk factors (recent antibiotics, diabetes, previous infection).
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the seminal vesicle–prostate junction.

Technique

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density >0.15 ng/mL/cc suggests cancer).
- 18-gauge spring-loaded needle; touch the capsule, withdraw slightly, then fire, so the core samples peripheral zone.
- **12 cores**: base, mid and apex on each side, medial and lateral (sextant missed lateral cancers). Add MRI-targeted cores (cognitive or fusion).

Complications

Complication	Rate
Haematuria	23–63%
Haematospermia	10–50%
Rectal bleeding	2–22%
Infection	0.1–7%; sepsis 0.3–3%
Retention	0.2–0.4%
Vasovagal reaction	1–5%

- Sepsis from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return. Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

Recent advances

- **MRI first:** PROMIS (MRI sensitivity 93% vs 48% for TRUS biopsy); PRECISION (targeted biopsy found more significant cancer, 38% vs 26%, and avoided 28% of biopsies).
- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU (European Association of Urology) 2023.

Diagram to draw: 12-core template on a transverse prostate outline.

Key points to remember:

- 12 cores; antibiotics ≤24 hours for all.
- Bleeding commonest; sepsis most serious.
- MRI before biopsy; transperineal now preferred.

Q3e. Management of valve bladder.

Type: SAQ (10 marks)

Definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV (posterior urethral valve) ablation, without residual obstruction, because of bladder dysfunction.

Pathophysiology (vicious cycle)

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Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual
+ polyuria (nephrogenic diabetes insipidus) → bladder never empty
→ pressure transmitted to dilated upper tracts → they refill the bladder → renal damage
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- Three processes: **polyuria, poor sensation, residual urine.**
- Age pattern: hyperreflexia in infancy → better compliance in childhood → large atonic bladder (myogenic failure) after puberty.

Clinical features

- Late continence (about 5.5 years), incontinence, UTI (urinary tract infection), worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**

Evaluation

- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

Management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones).

- Reflux is treated through the bladder; reimplantation is rarely needed. Manage CKD (chronic kidney disease).

Transplantation

- ESRD (end-stage renal disease) in 20–50%. Optimise the bladder first (videourodynamics, CIC or overnight drainage); graft survival is then similar.

Diagram to draw: vicious-cycle diagram of the valve bladder.

Key points to remember:

- Polyuria, poor sensation, residual urine.
- Overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Q3f. TOT.

Type: SAQ (10 marks)

Assumed meaning: TOT = transobturator tape, a mid-urethral sling for female stress urinary incontinence.

Introduction

- A synthetic **MUS (mid-urethral sling)** of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) described it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory:** pubourethral ligaments, suburethral hammock and pubococcygeus maintain continence.
- A loose tape at mid-urethra supports the posterior urethral wall, which compresses against it on straining. It anchors in the obturator muscles.

Indications and evaluation

- SUI (stress urinary incontinence) or stress-predominant mixed incontinence after failed pelvic floor training.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh and alternatives (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

Technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.
2. Stab incision in the labiocrural fold, at the notch between inferior pubic ramus and adductor longus tendon.
3. Trocar perforates the obturator membrane, turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornix.
5. Tension-free: a clamp passes easily between tape and urethra.

- **Inside-out** (vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- **TOMUS (2010):** objective cure 77.7% vs 80.8% retropubic (not different). Cochrane review (2017): similar cure beyond 5 years.
- Poorer with **ISD** (intrinsic sphincter deficiency) or a **fixed urethra**: retropubic sling preferred.

	TOT	Retropubic
Bladder perforation	0.6%	4.5%
Groin pain	6.4%	1.3%
Voiding dysfunction, bleeding	Less	More
Mesh exposure	about 2%	about 2%

Complications

- Groin pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- **Also know:** single-incision mini-slings; alternatives: colposuspension, autologous fascial sling, bulking agents.

Diagram to draw: pelvis with tape under mid-urethra through both obturator foramina.

> **Key points to remember:**
> - Delorme 2001; fewer bladder injuries than retropubic.
> - TOMUS: cure equivalent (77.7% vs 80.8%).
> - More groin pain; worse in ISD.

Rapid Revision Sheet – 2020-Summer Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe assessment and recent concept in management of metastatic RCC. (LAQ)

- IMDC: Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.

Q2. Discuss surgical management of erectile dysfunction. (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Infected implant: remove all parts; Mulcahy salvage 84%.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Q3a. CRPC in Ca prostate. (SAQ)

- Testosterone <50 ng/dL with PSA or radiological progression.
- AR still drives CRPC; keep ADT going.
- nmCRPC: apalutamide, enzalutamide, darolutamide.
- mCRPC: ARPI, taxanes, PARP inhibitor, radium-223, ^{177}Lu-PSMA.

Q3b. Retractile testis? (SAQ)

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

Q3c. Pre operative management of pheochromocytoma. (SAQ)

- α-blockade 7–14 days for all.
- Never β before α.
- Volume expansion prevents hypotension; watch glucose.

Q3d. TRUS biopsy. (SAQ)

- 12 cores; antibiotics ≤24 hours for all.
- Bleeding commonest; sepsis most serious.
- MRI before biopsy; transperineal now preferred.

Q3e. Management of valve bladder. (SAQ)

- Polyuria, poor sensation, residual urine.
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